

From: [Yarger, Alexandra M](#)
To: [robotics-phds-postdocs](#), [robotics](#)
Subject: Robotics Seminar Next Week - Thursday, August 27th, 15:30-16:30 - RSM 131
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Attachments: [Outlook-nkks1d3y.png](#)
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Dear All,

Dipankar Bhattacharya will be hosting **Simone Mentasti** and **Nicolò Botteghi** for a seminar next week.

Details are below. I hope to see you there!

When: **Thursday, August 27th, 15:30-16:30**

Where: **RSM 131**

Nicolò Botteghi:



Talk title: From Optimal Control to Reinforcement Learning: Opportunities and Challenges in Sequential Decision-Making.

Abstract: Optimal control and reinforcement learning share a common objective: the design of decision-making policies that optimize long-term performance objectives in dynamical systems. Despite their different historical origins – with optimal control rooted in mathematics and physics, and reinforcement learning in machine learning and artificial intelligence – their foundations are deeply interconnected. In this talk, starting from the optimal-control formulation, I will introduce the building blocks of reinforcement learning and explore the similarities and differences of the two approaches. Then, I will present a few approaches that I have recently developed to mitigate limitations of state-of-the-art reinforcement learning algorithms, namely (I) data inefficiency, (II) poor generalization to variations of the system dynamics, (III) limited robustness in presence of uncertainties and (IV) scalability to high-dimensional control problems. I will show applications of these reinforcement learning-based techniques to control problems frequently arising in engineering and applied science such as, for example, flow and density control, and navigation problems in unsteady flows.

Bio: I am Nicolò Botteghi and I am a postdoctoral researcher at the MOX laboratory in the Department of Mathematics at the Politecnico di Milano (<https://www.nicolobotteghi.com/>). I obtained my bachelor's and master's degrees in Automation Engineering at the University of Bologna and my PhD in Robotics and Mechatronics at the University of Twente in the Netherlands. My PhD research focused on reinforcement learning and dimensionality reduction for autonomous robotics. After receiving my PhD, I continued as a postdoctoral researcher in the Department of Applied Mathematics at the University of Twente, where I further developed my PhD research on reinforcement learning in the context of dynamical systems, and

scientific computing. My current research focuses on developing data-efficient, robust, and interpretable reinforcement learning approaches for dynamical systems. I'm particularly interested in the developments of data-driven control methods that can tackle real-world challenges at the intersection of flow control, robotics, and scientific machine learning. My expertise spans reinforcement learning, control of dynamical systems, model-order reduction, uncertainty quantification, and robotics.

Simone Mentasti:



Talk title: Multi-Sensor Perception for Autonomous Systems

Abstract: Reliable autonomy depends on the ability to transform incomplete and noisy sensor measurements into an accurate representation of the environment and of the system's own state. In this talk, I will present a selection of my research activities in perception for autonomous vehicles, with a particular focus on computer vision and its integration with radar and LiDAR sensing.

The talk will cover perception problems at different levels, from extracting geometric and semantic information from individual sensor observations to constructing representations that can support navigation and decision-making. I will discuss vision-based mapping and line detection, both online and offline, for real time navigation and mapping. I will then present approaches for vehicle state estimation, where visual information is combined with other sensing modalities to estimate motion and compensate for the limitations of individual sensors. Finally, I will address obstacle detection and tracking using cameras, radar, and LiDAR, highlighting the complementary properties of these modalities and the challenges involved in combining their measurements over time.

The talk will conclude with a broader perspective on how multi-sensor and distributed perception can move beyond isolated detection tasks toward interconnected autonomous systems, in which vehicles and intelligent infrastructure cooperate to build more accurate representations of the environment and improve overall system safety and robustness.

Bio: Simone Mentasti is a researcher at the Artificial Intelligence and Robotics Laboratory (AIRLab) of the Department of Electronics, Information and Bioengineering (DEIB), Politecnico di Milano. He received his M.S. degree in Computer Science from Università degli Studi di Milano in 2017 and his Ph.D. in Information Technology from Politecnico di Milano in 2022. His research interests lie in robotics and autonomous systems, with a particular focus on perception and sensor fusion. His work explores the development and application of robust perception methods across a variety of domains, including autonomous vehicles, indoor robotics, agricultural robotics, and smart eyewear. He also collaborates with the Department of Mechanical Engineering on research activities related to intelligent transportation, urban mobility, and autonomous driving. His collaboration with the Smart Eyewear Lab focuses on context awareness and eye tracking.

